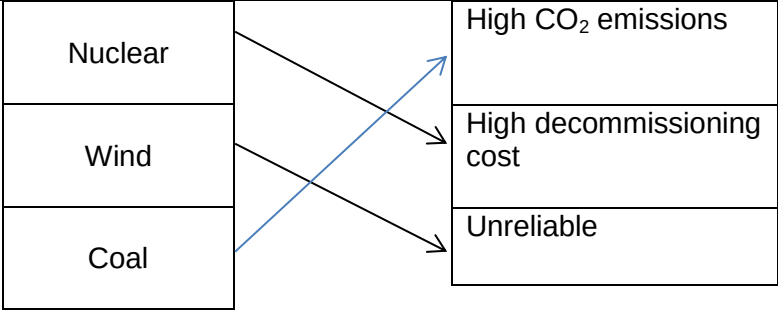


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4 FT	(a)	(i)		Nuclear Wind Coal High CO₂ emissions High decommissioning cost Unreliable  3 correct = 2 marks 1 or 2 correct = 1 mark More than one line drawn from a method don't credit	2			2		
		(ii)		Coal efficiency = $\frac{8}{20} \times 100$ (1) = 40% and disagree or it is the same (1) For an answer of 0.4 so agree or it is less efficient award 1 mark only Efficiency >100% with any comment gets zero Alternative: $\frac{40}{100} \times 20$ [MJ] (1) = 8 [MJ] and disagree or it is the same (1) Alternative: Wasted from coal: $\frac{12}{20} \times 100 = 60\%$ (1) Wasted energy nuclear power station = 60% and disagree or it is the same (1)			2	2	2	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		It is reliable [supply] / maintains supply in case of breakdown / responds to changing demand / has back-ups in case of demand or breakdown / it can import electricity from abroad if needed / works 24 hours a day Don't accept reference to efficiency	1			1		
		(ii)		A	1			1		
		(iii)		To increase voltage / step-up voltage / needs to be a high voltage / reduce current / needs to be a small current (1) Accept increase the volts or kV / decrease the amps Don't accept to increase the voltage and the current To reduce energy or heat loss / increase efficiency / reduce power losses [in lines] (1) Don't accept reference to increasing power or electricity or no energy loss	2			2		
				Question 4 total	6	0	2	8	2	0